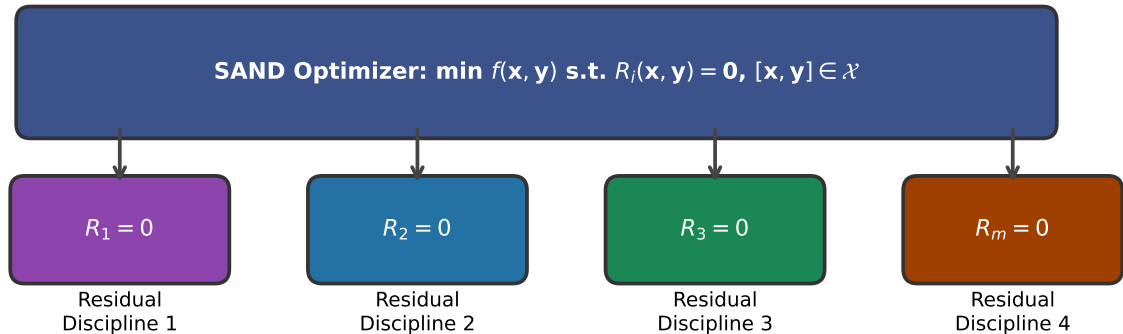


Simultaneous Analysis and Design (SAND) Architecture



All design $\mathbf{x}$ and response $\mathbf{y}$ variables passed simultaneously to optimizer

Residuals $R_i(\mathbf{x}, \mathbf{y}) = \mathbf{0}$ enforced as equality constraints

Enables gradient-based methods; disciplinary analyses run once per iteration